

PROBLEM STATEMENT #08 | Campus & Academic Operations

Digital Campus Library Reservation Gateway

1. Problem Context & Overview

A modern university library system to search physical ISBN catalogs, manage dynamic holding queues for borrowed copies, automate holding notifications, and track overdue fine transactions.

Target Stakeholders / Actors: *Student Member, Head Librarian*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall allow members to search the ISBN repository and place hold requests on currently loaned books to enter a FIFO waitlist.

Sample Acceptance Criteria: Pass: User is assigned correct position in queue; Fail: User already holding max allowed books places additional reservation.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The catalog search query shall return paginated results for 50,000 titles in less than 150 ms.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.